Weighted Learnable Recursive Aggregation Network for Visible Remote Sensing Image Detection

Xuehu Duan, Zihao Li, Jun Zhang, Shuohao Li, Jun Lei, and Lixin Zhan

***Abstract*— With the continuous development of intelligent autonomous aerial vehicles (AAV) technology, efficient and accurate sensing of surrounding objects through onboard sensors has become an important research direction. Among these, the object detection is one of the most common perception techniques, but generic object detection methods have low-detection performance in remote sensing images. To address this, we propose the weighted learnable recursive aggregation detection framework, which aims to improve the detection performance in AAV remote sensing images. The network maintains efficiency while ensuring the accuracy of detection. First, to improve the fusion capability of multiscale characterization of small objects, we design a multichannel weights learnable recursive aggregation network. The network improves the multiscale representation fusion capability by dynamically fusing different layers of scale features while recursively aggregating different layers of features. In addition, we design a multichannel recursive residual fusion mechanism for this framework, which is capable of extracting object feature information at different spatial scales and enhances the ability of multiscale characterization of the object. Then, we migrate the multichannel weights learnable recursive aggregation network to different detection frameworks to verify its generalization. Finally, we perform experimental validation using VisDrone 2019, AI-TOD dataset and comparing it with existing methods. The proposed model improves the mAP@0.5 value by 2.1% over the benchmark model on the VisDrone 2019 dataset. On the AI-TOD dataset, our proposed algorithm improves the mAP@0.5 by 1.7% compared with the benchmark model. Also working on the LLVIP dataset, our algorithm achieves 89.8%.**

***Index Terms*— 2-D object detection, feature fusion, multiscale, remote sensing image (RSI), autonomous aerial vehicles (AAV).**

I. Introduction

With the acceleration of urbanization and increasing traffic flow, the safety management of roads and public places is facing increasing challenges. The traditional way of police force patrol is restricted by the ground perspective and human resources, making it hard to realize real time and comprehensive monitoring of transportation and public places. Therefore, using drones for daily inspections has become a new solution. Autonomous aerial vehicles (AAVs) have technical advantages, such as high-altitude vision [1], high mobility, and object detection, enabling them to overlook the entire transportation network and public places from the air and achieve comprehensive monitoring without non-line-of-sight spots.

The AAV platform can achieve autonomous perception capability of the surrounding environment through onboard sensors and computer vision technology [2], [3]. One of the most commonly used environmental perception techniques is object detection [4], [5]. In recent years, the dynamic development in the field of deep learning [6] has brought significant advances and innovations in object detection techniques. Nevertheless, there are still many challenges and limitations in applying object detection techniques to AAV platforms. In this research work, we analyze the AAV datasets [7] and find that the main causes of the low-detection accuracy of the object detection model are as follows.

1) *Less Features*: In AAV images, the objects to be detected are dominated by small-sized objects (MS COCO datasets define objects with pixel values less than 32×32) [8]. Compared with large and medium scale objects, small-sized objects have very few visualized features in the image. In addition, most of today's generalized object detection models are adapted to detection scenarios with normal-sized objects and balanced samples. Therefore, conventional detectors cannot extract small-object features well.
2) *Category Similarity*: Fewer character traits of the objects to be detected make it difficult to distinguish between similar categories [9]. Changes in AAV view angle and altitude lead to the phenomena of a small percentage of object pixels, unbalanced scale, and unbalanced spatial distribution [10]. The detection models are unable to extract discriminative features from a small number of informative features. This leads to the problem of difficulty in differentiating between similar categories by the object detection models.
3) *Imbalanced Samples*: The sample datasets to be detected exhibit significant long-tailed distribution [11] characteristics. Most of the object detection models currently available use the balanced training strategy to achieve the ideal state. When facing the long-tailed distribution data, their generalization ability appears to be relatively weak.

The proposed method, although validated on VisDrone 2019, AI-TOD, and LLVIP datasets separately, takes into account that the VisDrone 2019 dataset contains more challenging scenarios. Therefore, this article focuses on analyzing the VisDrone 2019-DET dataset [12] and experimentally verifies the validity of the model on this dataset. The dataset is an open-source project initiated by the Department

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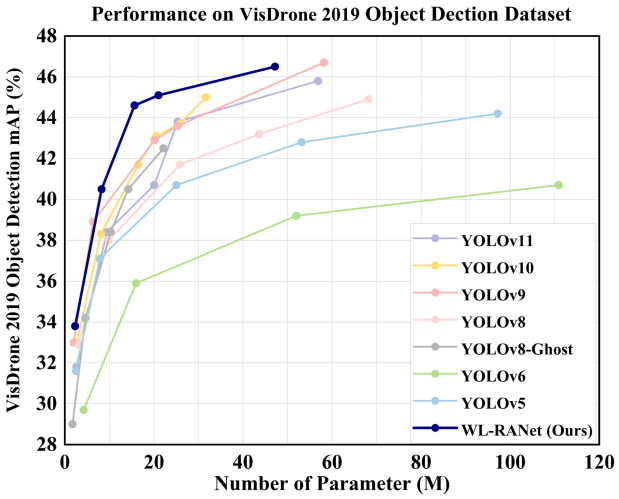


Fig. 1. Comparison of the performance of real-time object detection models on the VisDrone 2019 dataset, the folds of each model in this figure represent different sizes of that model, and the model results should be better closer to the left and above.

of Computer Science and Technology, Tsinghua University. It focuses on AAV vision object detection and tracking, aiming to promote research and development in the field of AAV vision. The dataset contains a large number of AAV-captured images and videos covering ten categories of objects in a variety of complex environments and viewpoints, such as pedestrians and vehicles. These images and video data cover a variety of locations (e.g., 14 different cities thousands of kilometers apart), environments (urban and rural), objects (pedestrians, vehicles, and bicycles), and densities (sparse and crowded scenes).

Aiming at the problems of low accuracy of object detection models due to small-object size, large-scale variation, and interclass similarity [5], [13], [14], we propose the weighted learnable recursive aggregation detection algorithm (WL-RANet) for AAV remote sensing image detection based on the YOLOv11 [15] framework. A comparison of the validation results of our proposed model with mainstream models on VisDrone 2019 data is shown in Fig. 1. The WL-RANet algorithm reconfigures the feature fusion part through a multichannel weights learnable recursive aggregation (McWL) network to improve the object multiscale fusion capability. In addition, we design a multichannel recursive residual fusion (McRF) mechanism to mine the deep representation information to solve the problem of low accuracy of similar category detection, and the main contributions include the following.

1) *McWL Network*: In this article, a weighted learnable recursive aggregation network is designed. It enhances the multiscale feature fusion capability of the network by dynamically fusing different layers of features and recursively aggregating the features at different levels. The detection accuracy of the model is guaranteed without affecting the inference speed of the network.
2) *McRF Mechanism*: To further enhance the small-object representation mining capability, we design a multichannel recursive residual fusion mechanism, which can efficiently extract the representation information on different scale spaces, and as a result, the model's ability to characterize the object at multiple scales is enhanced.
3) *Migration Generalization Validation*: We migrate the idea of the McWL network to different detection frameworks and verify the generalization and effectiveness of different detection frameworks in AAV remote sensing image detection tasks.
4) *Validation of Model Generalization*: Robustness and generalization of the proposed model are validated on VisDrone 2019, AI-TOD, and LLVIP datasets. Meanwhile, the performance of the proposed model is comprehensively compared and analyzed with the benchmark model on these three datasets.

The rest of this article is organized as follows. In Section II, various algorithms for AAV object detection are presented, and the AAV small-object detection algorithms and the long-tailed data object detection algorithms are presented, respectively. In Section III, a more suitable AAV object detection algorithm is proposed. In Section IV, we experimentally verify the feasibility of the proposed algorithm. In Section V, we summarize an outlook of this article.

II. Related Works

The AAV object detection is realized by sensing the surrounding environment through the onboard camera to detect the ground objects. Currently, vision-based object detection algorithms are relatively mature but still face some challenges in specific scenarios of AAV object detection. For example, phenomena, such as long-tailed distributions, low illumination, small-sized objects, and uneven scale and spatial distributions of objects pose difficulties for object detection algorithms and lead to poor robustness of detection models. In response to the above problems, relevant scholars have proposed a variety of effective ways to deal with them. Deep learning-based AAV object detection algorithms have currently become a mainstream approach, and these algorithms can be broadly categorized into two types of frameworks: two stage and one stage. In this section, we focus on how researchers have tackled the challenge of degrading model detection accuracy in aerial images due to phenomena such as long-tailed distributions and small-sized objects.

A. Small-Sized Object

In AAV aerial images, small-sized objects occupy a dominant position. Therefore, for such complex aerial images, how to quickly and accurately detect the small objects in them not only has far-reaching theoretical significance but also shows great value in practical applications. As an important branch of generalized object detection, the small-object detection has long been paid close attention to by researchers. Existing methods for small-object detection mainly encompass techniques in feature extraction and enhancement, improvements in network architecture, as well as innovations in loss functions and training strategies.

To enhance the object localization capability of detection models, Liu et al. [16] proposed a multibranch parallel feature pyramid network (FPN), which efficiently extracts richer and more detailed features through a cascaded architecture while maintaining a smaller model size. Similarly, Wang et al. [17]

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improved the double-head R-CNN by integrating transformer and deformable convolution, designing a content-aware feature reorganization FPN and reconfiguring anchor box parameters, thereby effectively enhancing the feature extraction capability of small objects in AAV aerial images.

For improving small-object detection performance, Liang et al. [18] proposed a feature fusion and scale detector based on spatial context analysis, which further optimizes the SSD and enhances the detection of small objects in AAV images. Meanwhile, a feature pyramid is constructed by introducing additional scale branches, which are combined with the features generated and fused by the inverse convolutional block to realize joint prediction. To prevent the missing object information, Zheng and Chen [19] designed a bidirectional stepped cascade feature pyramid method based on SSD. The method effectively avoids the problem of information missing during feature transfer through step cascading, and its bidirectional design ensures that the fused features retain both rich detail information and high-level semantic information. In addition, Zheng and Chen [19] proposed an attentional interaction module that can aggregate features from different streams more efficiently, thus further improving the performance of small-object detection.

One of the main reasons for the degradation of small-object detection performance is the inadequacy of the model to represent small-scale features. To this end, Shamsolmoali et al. [20] constructed an image pyramid detector by combining an image pyramid network with SSD. The detector is initially designed to enhance the capability of FPNs for small-scale feature representation to improve the detection performance of small objects in AAV aerial images. To make the object detection features to be detected more significant, [21] increases the feature map channel information by introducing convolutional blocks between the backbone network and the neck based on YOLOv5 and reduces the computational cost by employing the attention mechanism.

In small-object detection scenarios, optimization in feature representation is not only needed, but the optimization of the network structure is also crucial. Liu et al. [22] have optimized the network structure based on the YOLOv3 model. In [22], the residual block in Darknet is improved by connecting two residual units with the same width and height. In addition, to enrich the spatial information in the early layers of the network, the authors add a convolution operation in the early layers, thus optimizing the whole network structure. In order to fuse the feature information more effectively, Liu et al. [23] proposed an enhanced spatial pyramid pool structure, which realizes the deep fusion of feature information through the self-feature expansion module. In [24], the Ghost bottleneck C3 module with Ghost convolution module is introduced, which aims to capture richer feature information and significantly improve the operation efficiency of the model. The detection of small-object features is enhanced by a well-designed weighted bidirectional feature pyramid structure combined with the convolutional attention module.

Concerning the training strategy, the traditional model is mainly oriented to train on regular object datasets, which is not applicable to the difficult-to-learn small-object representations. In this regard, Huang et al. [25] proposed a feature-guided enhancement module and scale-aware weighted loss (SAWL), which significantly enhances the contribution of small objects in training by dynamically adjusting the loss weights.

To solve the problem of degradation of generalization ability due to domain offset in AAV object detection, the researchers find experimentally that the difference in the contribution of different frequency bands to the generalization ability of AAV object detection is more significant than that of general object detection. Therefore, Wang et al. [26] solved the problem of degradation of generalization ability due to domain offset in AAV object detection by extracting domain-invariant and domain-specific spectra through a frequency-domain decoupling framework. Aiming at the problem of high density of small objects in AAV images, Yingwei and Xiaoxia [27] optimized the YOLOv6 object detection network by mixing the data enhancement and feature alignment modules to improve the accuracy of object detection in high-density scenes.

B. Long-Tailed Distribution

By analyzing the distribution of AAV data, we find that the data has a significant long-tailed distribution. It is also one of the main reasons for the decrease in model detection accuracy. Gao et al. [11] proposed a new long-tailed loss function, PGLoss, and a parallel block for object detection in remote sensing images (RSIs). PGLoss solves the problem of long-tailed distribution in a one-stage network and controls false alarms by dynamically balancing and assigning weights based on the frequencies of each category in the dataset. Parallel block balances accuracy and inference speed through residual optimization and structural reparameterization.

Focusing on the problem of long-tailed object detection, Zhao et al. [28] pointed out that existing detectors suffer from the problem of imbalance between label sampling and foreground–background. The authors propose the logit normalization (LogN) method, a simple and effective a posteriori calibration method for long-tailed object detection. This article demonstrates the effectiveness of LogN by analyzing the relationship between the logit statistic and the resampling training label distribution, as well as the independence of the background margin from the label distribution. The authors verify the superior performance and generalization ability of LogN, which outperforms existing methods in a variety of settings and performs well on different types of detectors, through extensive experiments, as well as a detailed ablation study and analysis.

III. Methods

A. WL-RANet Framework

To effectively solve the problem of poor model robustness and difficult generalization caused by the phenomena of small-object size, wide-scale variation in AAV remote sensing images, we propose the weighted learnable recursive aggregation network detection model, and its network structure is shown in Fig. 2. The proposed algorithm primarily consists of three key components: feature extraction, feature fusion, and prediction mechanism.

Initially, the backbone network consists of selective channel downsampling (SCD), McRF network, spatial pyramid pooling-fast (SPPF), and C2f-Pointwise spatial attention

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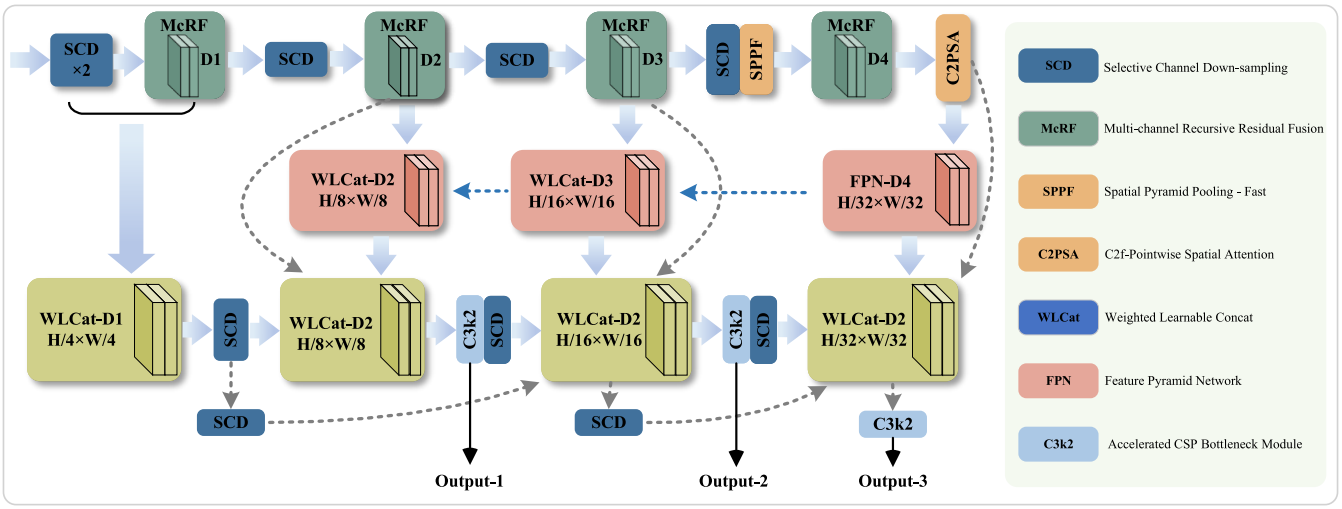


Fig. 2. WL-RANet detection model network architecture.

(C2PSA), which is mainly used for enhanced object feature extraction.

WL-RANet backbone network mainly solves the problem of incomplete extraction of object features by the network due to the small-object size in AAV remote sensing images. It utilizes selective channel downsampling and multichannel recursive residual fusion mechanism to mine the object detail information from different layers of the original image to make up for the phenomenon of small-sized object information loss, and then solve the problem of small-object detection performance degradation.

Second, a feature fusion network is used for information fusion processing. The McWL network solves the problem of reduced detection accuracy due to the uncertainty of the object angle and wide variation of the scale. It mainly utilizes the multidimensional weights learnable fusion mechanism, and the recursive aggregation network strategy to enhance the feature representation at different scales for the high-level object semantic information and the low-level semantic information. Finally, the object location and class information are regressed to the output using the prediction mechanism.

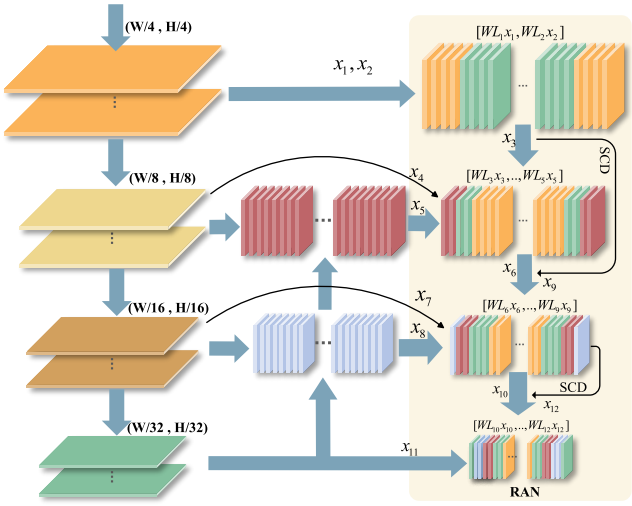


Fig. 3. McWL network structure.

B. McWL Network

In generalized object detection models (e.g., YOLO series), the feature fusion strategy is to improve the adaptability of the model to multiscale variations by fusing shallow features with deeper features. Among these, the superficial features have higher resolution and contain more small-object location and detail information, but due to less feature extraction, they result in lower semanticity and more noise. Deep features, on the other hand, are less capable of perceiving details, and even small-object information are blurred out. In order to synthesize superficial and deep information, research scholars have proposed a feature fusion network combining FPN and path aggregation network (PAN) [29], [30]. We find through our study that the feature mining of PAN networks is too aggressive, which may lead to poor performance of small objects on the feature map. In addition, the PAN network has a simple channel concatenation for the fusion of the underlying features with the higher level features in the feature fusion process. This approach is more homogeneous and does not measure the importance of each channel feature to the network, making small-object information lost. To address the above problems, we propose the multichannel weights learnable recursive aggregation network, as shown in Fig. 3.

As can be seen from Fig. 3, the network aggregates multichannel weights learnable and feature recursion ideas. First, in order to be able to effectively capture the information of objects at different scales, we design a bidirectional propagation path in the feature extraction part. Simultaneous delivery of feature representations at different scales to FPN and RAN networks for multichannel feature fusion at different layers. The method enables the network to adapt to multiple channel feature changes during feature fusion and improves the performance of the model in complex scenes with small objects. In addition to this, feature fusion in the usage-based model is simply splicing in dimensions and is unable to learn the importance of the feature maps in each channel through spontaneous learning. To this end, we design an adaptive multichannel weights learnable feature fusion mechanism. The

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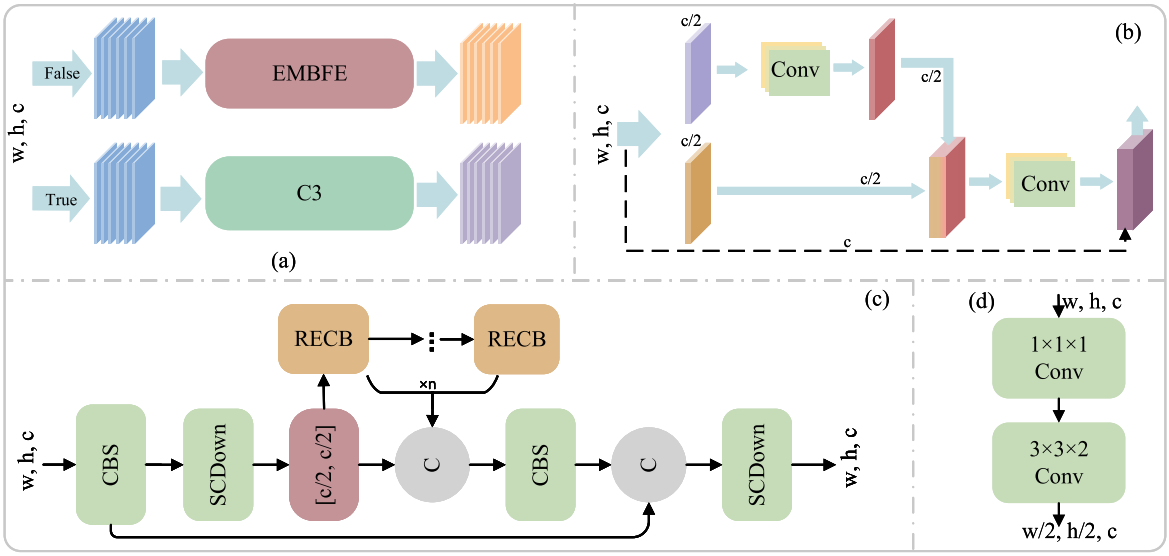


Fig. 4. Structure of the multichannel recursive residual fusion mechanism. (a) Multichannel recursive residual fusion mechanism. (b) Recursive cross-stage bottleneck. (c) Efficient multibranch feature extraction. (d) SCDown.

mechanism works by learning the importance of each channel feature to the network, and its mathematical expression is represented as (1)–(3)

$$\text{Weight} = \frac{w}{\sum_{i=0}^{c} wi + \varepsilon} \tag{1}$$

where w is the initial weight vector, w_i denotes each weight element within w, and ε represents a constant added to prevent division by 0 during the normalization process. This ensures numerical stability when normalizing the weights. Following the weight normalization, a weighted operation is applied to the features of each channel. This operation is expressed by the following equation:

$$\tilde{x}_c = \text{Weight}_c \bullet x_c. \tag{2}$$

Here, x_c represents the feature from the cth channel and Weight_c is the normalized weight corresponding to that channel. This weighting mechanism allows the model to emphasize or de-emphasize features from different channels based on their importance. After applying the weights to the channel features, the next step involves concatenating these weighted features along the channel dimension. This concatenation operation is described by the following equation:

$$\text{Output} = \text{Concat}([x_1, \ldots, x_c], d = 1), \quad c \in 1, 2, 3 \tag{3}$$

where x_c represents the weighted features from different channels and c indicates the number of channels. The concatenation operation combines these features along the specified dimension, facilitating the integration of multichannel information.

Then, a recursive aggregation network of features is designed. This structure not only alleviates the phenomenon of over-aggressive downsampling in PAN networks but also better aggregates feature information at different levels and enhances the fusion of low-level details with high-level semantic features. In addition, this network structure has smoother transmission between different layers, which reduces certain information loss and improves the overall detection accuracy of the model.

C. McRF Mechanism

For small objects in remote sensing images, it is also challenging to extract effective representations of small objects in complex backgrounds. In the generic object detection framework, the image data undergo a feature extraction network to first extract the object representations for the feature fusion stage, thus the model performance is inextricably linked to the feature extraction network. We design a multichannel recursive residual fusion mechanism for AAV remote sensing image detection scenarios. As shown in Fig. 4, the mechanism mainly consists of an efficient multibranch feature extraction module and a recursive cross-stage bottleneck.

The WL-RANet detection model feature extraction network continues the CSP-Darknet network idea using convolutional layers to downsample remote sensing image data. This module is the basis of the in-depth extracted representation of the network, where the image representation gradually decreases in spatial dimensions as the convolutional layers are stacked, while increasing the number of feature channels. The significant improvement in the feature extraction network of the WL-RANet detection model is the introduction of a multichannel recursive residual fusion mechanism. The structure of the mechanism is shown in Fig. 4(a), which constructs a multichannel recursive bottleneck. The problem of reduced model accuracy due to the lack of information on small objects in the image is solved by increasing shallow representation and deep representation through broadening the network width and multilevel perceptual fusion. From Fig. 4(a), it can be seen that when the bool value of the McRF input is "true," the features go through the default C3 structure for deep feature extraction. When it is "false" for the bool value, the feature map then passes through our proposed efficient multibranch feature extraction (EMBFE) module, which is able to perform feature extraction at three different levels.

The global perception of the model is enhanced by designing a recursive cross-stage bottleneck (RECB) that fuses contextual information in the feature map at multiple levels. This module enriches cross-layer information exchange by

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fusing feature layers from different layers to help capture information from different layers and avoid over-reliance on information from one layer. Its structure is shown in Fig. 4(b). Its mathematical expression can be written as follows:

$$x_1, x_2 = \text{split}(x, 2, \text{axis} = 1) \tag{4}$$
$$Y_1 = f_{\text{conv}}(x_1) \tag{5}$$
$$Y_2 = f_{\text{conv}}(Y_1) \tag{6}$$
$$Y_3 = f_{\text{conv}}(\text{Cat}(Y_2, x_2)) \tag{7}$$
$$Y_4 = f_{\text{conv}}(Y_3) \tag{8}$$
$$Y = \begin{cases} x \oplus Y_4, & \text{if self.add and } C_1 = C_2 \\ Y_4, & \text{otherwise.} \end{cases} \tag{9}$$

Equations (4)–(9) describe the mathematical formulation of the RECB module. x represents the input feature, f_{conv} denotes the convolution operation, Cat indicates the feature map concatenation, Y_i represents the output, and $\oplus$ symbolizes elementwise feature summation.

Specifically, the input feature x is evenly split along the channel dimension (axis=1) into two parts, x_1 and x_2. This splitting operation allows subsequent convolution operations to process different subsets of features separately. Then, the feature subsets after segmentation are subjected to four convolution operations to obtain the Y_4 features. Finally, If the conditions (self.add is true and $C_1 = C_2$) are met, the input feature x is elementwise added ($\oplus$) to Y_4, producing the final output Y. Otherwise, Y_4 is directly output.

To increase the width and depth of the feature extraction network, we have designed the EMBFE module, which can be seen from Fig. 4(c), which is mainly divided into three gradient streams for shallow, medium and deep feature extraction, respectively. Its mathematical expression for the forward reasoning process can be written as follows:

$$x_1 = f_{\text{conv}}(x) \tag{10}$$
$$y_1, y_2 = \text{split}(f_{\text{SCDown}}(x_1)) \tag{11}$$
$$y_n = \bigcup_{i=1}^{n} \text{RECB}(y_2) \tag{12}$$
$$x_2 = f_{\text{conv}}(\text{cat}(y_1, y_n)) \tag{13}$$
$$\hat{y} = f_{\text{SCDown}}(\text{cat}(x_2, x_1)). \tag{14}$$

f_{SCDown} denotes a selective channel downsampling operation, $\bigcup_{i=1}^{n}$ represents the concatenation of each bottleneck output, and $\hat{y}$ indicates the module's final output. Equations (10)–(14) describe the forward reasoning process of the EMBFE module.

The input feature x undergoes a convolution operation f_{conv}, generating the initial feature x_1. This step extracts low-level information from the input features. The initial feature x_1 undergoes selective channel downsampling f_{SCDown}, and the downsampled feature is split along the channel dimension into two parts, y_1 and y_2. Selective channel downsampling adaptively reduces the number of feature channels while preserving important information.

Thereafter, the split feature y_2 is processed by RECB modules, and the outputs are concatenated ($\bigcup_{i=1}^{n}$). It further extracts features through convolutional and nonlinear transformations and enhances the feature representation.

TABLE I
EXPERIMENTAL SOFTWARE AND HARDWARE PARAMETERS

Name	Parameters
System	Ubuntu 24.04
Framework	pytorch 2.4.0
Python	3.10.14
CUDA	12.1
GPU	NIVIDIA GeForceRTX 3060 12G
CPU	12th Gen Intel(R) Core(TM) i5-12400

Finally, the initial split feature y_1 and the recursively processed feature y_n are concatenated, and a convolution operation is applied to the concatenated features, generating feature x_2. The fused feature x_2 is concatenated with the initial feature x_1, and selective channel downsampling f_{SCDown} is applied to generate the final output $\hat{y}$. This module effectively extracts and fuses shallow, intermediate, and deep feature information through a multilevel gradient flow design, enhancing the model's feature representation capabilities.

In order to improve the computational efficiency as well as reduce the computational complexity of the feature extraction network, the selective channel downsampling strategy is introduced instead of the original CSP-Stage, the structure of which is shown in Fig. 4(d). Compared with the convolutional layer in the original feature extraction network, the selective channel downsampling strategy can better capture the target representation while maintaining a lower computational complexity. The strategy incorporates point-by-point and depth-by-depth convolution. To achieve computational lightness, the number of channels of the input tensor is linearly transformed by using point-by-point convolution. Each channel is then processed independently by depth-by-depth convolution, which maintains a constant number of channels while spatially downsampling the characterization data.

IV. EXPERIMENTAL ANALYSES

A. Experimental Environment and Settings

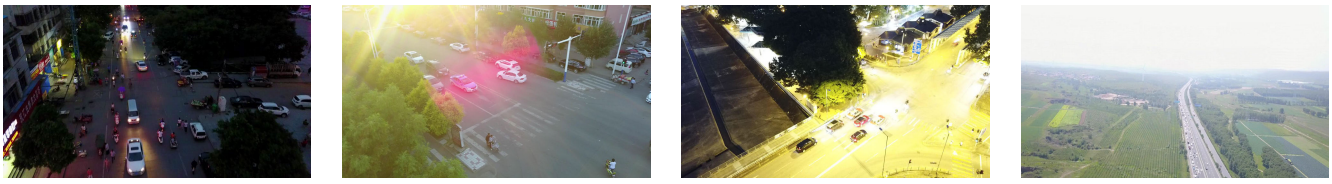
The state-of-the-art (SOTA) algorithm YOLOv11 is chosen as the baseline for the experiments and all experiments are performed on the VisDrone 2019, AI-TOD, and LLVIP public datasets of AAV remote sensing images. The experimental software and hardware environments are shown in Table I.

To address the issues of model overfitting and bias toward less sampled categories caused by long-tailed data during training, we employ various strategies, including data augmentation, category-balanced loss functions, and pretraining methods.

During the model training process, in order to solve the overfitting problem caused by the long-tailed data distribution, we introduce data augmentation techniques in the data preprocessing stage. The specific parameter settings are shown in Table II.

And then, our proposed model employs distribution focal loss (DFL) function. Its primary purpose is to solve the problem of class imbalance in object detection and to mitigate overfitting or bias problems in the long-tailed data, particularly

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(a) Evening (b) Backlighting (c) Overexposure (d) Misty day



(e) Cloudy weather (f) Small object (g) Weak light (h) Intersection

Fig. 5. Examples of typical challenges in VisDrone 2019, AI-TOD, and LLVIP datasets include (a)–(d) samples from the VisDrone 2019 dataset, (e) and (f) samples of the AI-TOD dataset, and (g) and (h) samples of the LLVIP dataset.

TABLE II
DATA AUGMENTATION PARAMETER SETTINGS

Methods	Parameters
HSV Hue	0.015
HSV Saturation	0.7
HSV Value	0.4
Translation	0.1
Zoom	0.5
Horizontal Flip Probability	0.5
Mosaic Augmentation	1.0
Copy-Paste Data	Flip
Random Erasing	0.4
Crop Ratio	1.0

for categories with fewer samples. The DFL function is combined with the traditional cross-entropy loss to enhance the model's ability to learn from long-tailed category data. The core idea of DFL function is to make the model focus more on hard-to-learn samples in long-tailed data by weighting the prediction probability of each category. Its mathematical expression is as follows:

$$\mathrm{DFL} = -\sum_{i=1}^{N}\sum_{c=1}^{C} y_{\mathrm{ic}}\left(\alpha(1-p_{\mathrm{ic}})^{\gamma}\log(p_{\mathrm{ic}}) + (1-\alpha)p_{\mathrm{ic}}^{\gamma}\log(1-p_{\mathrm{ic}})\right) \quad (15)$$

where N represents the number of samples, C denotes the number of categories, y_{ic} is the true label of the ith sample, p_{ic} is the predicted probability that the i sample belongs to a class c, α is a balancing factor used to adjust the weight between positive and negative samples, and γ is a focus parameter used to control the degree of attention to difficult samples. When γ is larger, the model pays more attention to the samples with lower predictive probabilities, thus improving the accuracy of the difficult samples. Finally, in the model training process, we use a transfer learning method with pretrained weight.

B. Analysis of Dataset

As shown in Fig. 5, some samples of VisDrone 2019, LLVIP, and AI-TOD datasets are presented, respectively. From what is presented, it can be clearly observed that each of these three datasets has its own characteristics, covering a variety of scenarios including diverse weather conditions, light environmental changes, and small objects.

1) VisDrone 2019: We set up the training set, the validation set, and the test set by pre-processing the data from the VisDrone 2019 dataset (https://github.com/VisDrone/VisDrone-Dataset) according to the official standards. The training set contains 6471 samples, the validation set contains 548 samples, and the test set is 3190 samples (1580 images for the challenge competition and 1610 for public algorithm evaluation). The number of categories, object sizes, and challenge difficulties in the dataset are also analyzed separately.

First, our study of the environment in which the dataset is located reveals that the dataset contains different weather (sunny, cloudy, and light haze), different scenes (urban and suburban), and different illumination (backlighting, daytime, and dark) conditions are collected. In addition to this different densities (congested and sparse) as well as objects (vehicles and pedestrians) are covered.

For the analysis of the number of categories, our study finds that the number of samples in the VisDrone 2019 dataset shows a long-tailed characteristic distribution, as shown in Fig. 6. It is easy to find that the data of "car," "pedestrian," and "people" appear more frequently, accounting for about 72.3% of the total sample. The seven categories of "motor," "van," "bicycle," "truck," "awning-triycle," and "bus" account for only 27.7% of the total sample. Due to the unbalanced distribution of sample data, the model is prone to favor high-frequency categories and ignore low-frequency categories, and it is prone to make the model difficult to learn and generalize.

For the object size analysis, we first define the small object as less than 32 × 32 pixels, the medium-sized object as 32 × 32.96 × 96 pixels, and the large size as more than 96 × 96 pixels, based on the COCO dataset. Subsequent data merging of training, validation, and testing data is performed for overall analysis. The object size in the original image and

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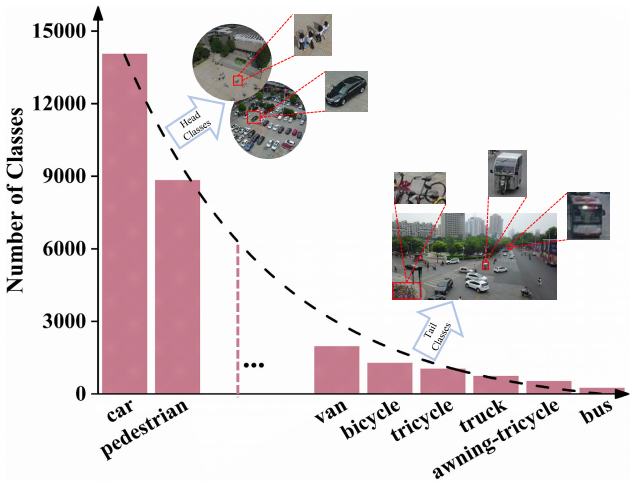


Fig. 6. VisDrone 2019 data characterization visualization.

TABLE III
VISDRONE 2019 OBJECT SIZE ANALYSIS

Datasets	Classes	Object Amount		
		Small	Medium	large
Original size (Pixel Range: 480×360 -2000×1500)	Pedestrian	91832	16934	421
	People	34166	4287	107
	Bicycle	9223	3711	135
	Car	91878	81302	13825
	Van	14600	15115	2987
	Truck	5050	8543	2691
	Tricycle	3063	3084	240
	Awning-tricycle	1944	2217	216
	Bus	2476	5086	1555
	Motor	30796	9302	280
	All	**285028**	**149581**	**22457**
Resized (Pixel: 640×640)	Pedestrian	106625	2548	14
	People	38107	445	8
	Bicycle	12558	510	1
	Car	146645	39251	1109
	Van	24908	7548	246
	Truck	10010	5750	524
	Tricycle	5213	1161	13
	Awning-tricycle	3552	812	13
	Bus	4993	3770	354
	Motor	39067	1306	5
	All	**391678**	**63101**	**2287**

the object size after resizing is also analyzed, respectively, and the results are shown in Table III.

As can be seen from Table III, small- and medium-sized objects account for 95% of all object sizes in the original size photograph. We consider that in the detection model framework, the image needs to be resized to 640×640, so we statistically analyze the object size data after resizing, and the small- and medium-sized objects accounted for 99.5% of all the size objects, and the small objects account for 85.7%.

TABLE IV
AI-TOD OBJECT SIZE ANALYSIS

Datasets	Classes	Object Amount		
		Small	Medium	large
Original size (Pixel Range: 480×360 -2000×1500)	Airplane	549	244	-
	Bridge	612	40	-
	Storage-tank	7315	431	-
	Ship	16484	846	-
	Swimming-pool	286	41	-
	Vehicle	302544	5402	-
	Person	17872	95	-
	Wind mill	243	-	-
	All	**345905**	**7099**	**-**
Resized (Pixel: 640×640)	Airplane	646	147	-
	Bridge	632	20	-
	Storage tank	7536	210	-
	Ship	16785	12	-
	Swimming pool	315	2389	-
	Vehicle	305557	5750	-
	Person	17925	42	-
	Wind mill	243	-	-
	All	**349639**	**3365**	**-**

2) AI-TOD: AI-TOD is a dataset designed for tiny object detection in aerial images (https://github.com/jwwangchn/AI-TOD) [31], [32]. The dataset contains 28 036 aerial images with a total of 700 621 object instances labeled, covering eight categories. Compared with existing aerial image object detection datasets, the average size of objects in AI-TOD is about 12.8 pixels. We have fully analyzed this dataset and the results are shown in Table IV. In the original image, the percentage of small objects is 97.9%. After image preprocessing, the percentage of small objects is 99.0%.

3) LLVIP: The LLVIP dataset is a visible and infrared image pairing dataset for low-light vision (https://github.com/bupt-ai-cz/LLVIP). The dataset contains 30 976 images, of which 15 488 pairs are visible and infrared image pairs [33]. Most of the images are taken in very low-light conditions and are mainly concerned with the detection and recognition of pedestrians. In this article, we only use the visible image part of the LLVIP dataset to verify the generalization ability of the model.

C. Comparison Between WL-RANet and SOTA Methods

To demonstrate the effectiveness of our algorithm, we generally compare it with the current SOTA models in the single-stage series. In Table V, we list the main components of the feature extraction networks and feature fusion networks for each model. It can be observed that the backbone networks in various models primarily consist of downsampling, residual modules, and lightweight modules. In the feature fusion networks, a combination of FPN and PAN is primarily adopted. In addition, we compare different versions of each model (representing models of different sizes and complexity,

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TABLE V
COMPARISON OF SOTA DETECTION MODELS ON VISDRONE 2019

Model	Version	Backbone	Neck	[Depth, Width, Max-channels]	mAP@0.5(%)	mAP@0.5:0.95(%)
YOLOv5 [34]	v5-n	Conv + C3 + SPPF	F-PANet	[0.33, 0.25, 1024]	31.6	18.3
	v5-s			[0.33, 0.50, 1024]	37.1	22.1
	v5-m			[0.67, 0.75, 1024]	40.7	24.6
	v5-l			[1.00, 1.00, 1024]	42.8	26.0
	v5-x			[1.33, 1.25, 1024]	44.2	27.3
YOLOv6 [35]	v6-n	Conv + RepVGG + SPPF	Rep-PANet	[0.33, 0.25, 1024]	29.7	17.4
	v6-s			[0.33, 0.50, 1024]	35.9	21.5
	v6-m			[0.67, 0.75, 768]	39.2	23.7
	v6-l			[1.00, 1.00, 512]	40.7	24.8
	v6-x			[1.00, 1.25, 512]	42.3	25.9
YOLOv8 [36]	v8-n	Conv + C2f + SPPF	F-PANet	[0.33, 0.25, 1024]	32.9	19.1
	v8-s			[0.33, 0.50, 1024]	38.0	22.9
	v8-m			[0.67, 0.75, 768]	41.7	25.4
	v8-l			[1.00, 1.00, 512]	43.2	26.5
	v8-x			[1.00, 1.25, 512]	44.9	27.6
YOLOv8-Ghost	v8g-n	GhostConv + C3Ghost + SPPF	F-PANet	[0.33, 0.25, 1024]	29.0	16.1
	v8g-s			[0.33, 0.50, 1024]	34.2	19.9
	v8g-m			[0.67, 0.75, 768]	38.4	22.7
	v8g-l			[1.00, 1.00, 512]	40.5	24.3
	v8g-x			[1.00, 1.25, 512]	42.5	25.5
YOLOv9 [37]	v9-t	AConv + RepNSCPELAN + SPPELAN	F-PANet	-	33.0	19.3
	v9-s			-	38.9	23.4
	v9-m			-	42.9	26.2
	v9-c			-	43.6	26.4
	v9-e			-	46.7	29.0
YOLOv10 [38]	v10-n	Conv + C2f + C2fCIB-SPPF-PSA	F-PANet	[0.33, 0.25, 1024]	33.0	19.2
	v10-s			[0.33, 0.50, 1024]	38.3	22.9
	v10-m			[0.67, 0.75, 768]	41.7	25.5
	v10-b			[0.67, 1.00, 512]	43.1	26.3
	v10-l			[1.00, 1.00, 512]	43.7	26.7
	v10-x			[1.00, 1.25, 512]	45.0	27.7
YOLOv11 [15]	v11-n	Conv + C3k2 + SPPF-C2PSA	F-PANet	[0.33, 0.25, 1024]	31.8	18.2
	v11-s			[0.50, 0.50, 1024]	38.4	22.9
	v11-m			[0.50, 1.00, 512]	43.0	26.3
	v11-l			[1.00, 1.00, 512]	43.8	26.9
	v11-x			[1.00, 1.50, 512]	45.8	28.5
WL-RANet	WL-n	SCDonv + McRF + SPPF-C2PSA	McWL-Net	[0.50, 0.25, 1024]	33.8↑	19.8↑
	WL-s			[0.50, 0.50, 1024]	40.5↑	24.4↑
	WL-m			[0.50, 1.00, 512]	44.6↑	27.6↑
	WL-l			[1.00, 1.00, 512]	45.1↑	28.0↑
	WL-x			[1.00, 1.50, 512]	46.5	29.0

with increasing network complexity from top to bottom in the table).

Based on the characteristics of objects in remote sensing images, we address the issues of reduced model accuracy caused by small size, wide-scale variation, and long-tail features, respectively. In Table V, first in backbone, we design a multichannel recursive residual fusion mechanism for being able to lighten the network as well as mining small object information in different layers of features, which contains a multilevel recursive fusion mechanism, a selective channel downsampling strategy, and a recursive cross-stage bottleneck. Second, in order to improve the multiscale representation fusion capability of the model, a dynamic multichannel weight learnable recursive aggregation network is designed inspired by the recursive idea.

We do this by quantitatively comparing with different versions of current SOTA and mainstream algorithms. We take the "s" version of each model as an example to compare the superiority of our model under the condition that the evaluation metric is mAP@0.5. Our model demonstrates improvements of 3.4%, 2.5%, 6.3%, and 2.1%, respectively, compared with the YOLOv5s, YOLOv8s, YOLOv8s-G (Lightweight version for YOLOv8s), and YOLOv11s models proposed by ultralytics [15], [34], [36]. Compared with the YOLOv6s detection model developed by Meituan's Vision Intelligence Department [35], our model demonstrates an improvement of 4.6%. Our detection model accuracy is improved by 2.2% and 1.6%, respectively, compared with YOLOv10s proposed by the Tsinghua Team and YOLOv9s proposed by Wang et al. [37], [38]. Among these, the visualization results under different weather and lighting conditions are shown in Figs. 7–9, which clearly shows the superiority of our algorithm.

In addition to this, we verify the generalization ability of the benchmark model with our proposed model on AI-TOD and LLVIP datasets, and the experimental comparison results are shown in Table VI. First, we migrate the model to the AI-TOD dataset for validation, and through the experiments, we find that our method improves the generalization ability of

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Fig. 7. Comparison of detection results between WL-RANet and mainstream algorithms in low-light environments. (a) YOLOv5. (b) YOLOv6. (c) YOLOv8. (d) YOLOv8-G. (e) YOLOv9. (f) YOLOv10. (g) YOLOv11. (h) WL-RANet.

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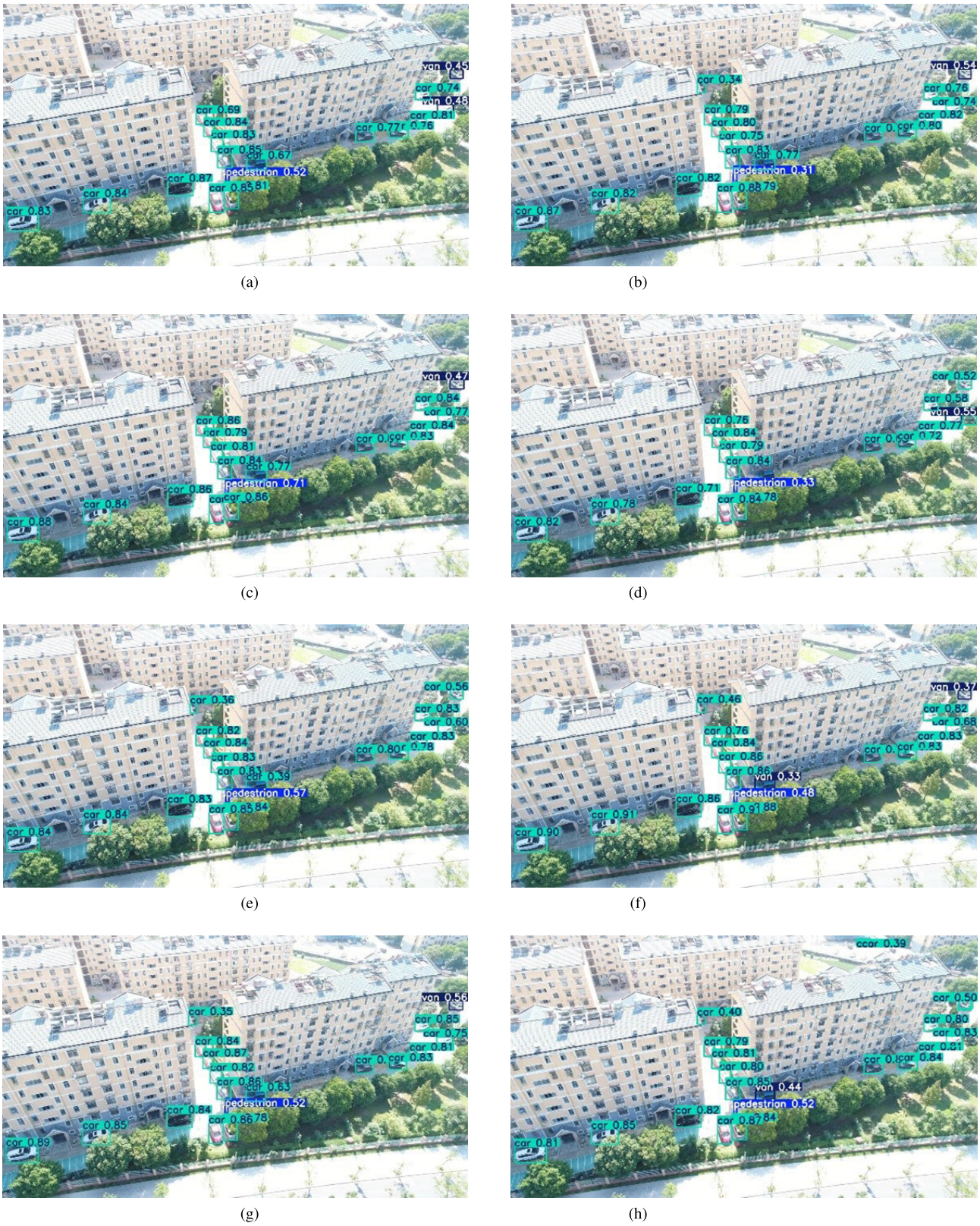


Fig. 8. Comparison of detection results between WL-RANet and mainstream algorithms in overexposed environments. (a) YOLOv5. (b) YOLOv6. (c) YOLOv8. (d) YOLOv8-G. (e) YOLOv9. (f) YOLOv10. (g) YOLOv11. (h) WL-RANet.

precision, recall, mAP@0.5, mAP@0.5:0.95 by 1.9%, 2.9%, 1.7%, and 1.2% than the YOLOv11 model, respectively. Second, it is validated on the LLVIP dataset and our model accuracy reaches 89.8% compared with mainstream models.

And then, we visualize the feature maps of different layers under the structure of the benchmark model and the WL-RANet model. Fig. 10 shows the results of the visualization of different frames corresponding to the deep output

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Fig. 9. Comparison of detection results between WL-RANet and mainstream algorithms in cloudy weather. (a) YOLOv5. (b) YOLOv6. (c) YOLOv8. (d) YOLOv8-G. (e) YOLOv9. (f) YOLOv10. (g) YOLOv11. (h) WL-RANet.

characteristics of the network. From the figure, we can see that in the C3k2 d2 and McRF d2 stages, our proposal McRF d2 can better extract the contour information of the target. Meanwhile, in the FPN d1 and McWL d1 phases, the effectiveness of our proposed McWL network is verified, and it can be clearly seen that our proposed method embodies more object characterization information. Finally, after the comparison of the output results of the network, the effectiveness

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TABLE VI
COMPARISON OF SOTA DETECTION MODELS ON AI-TOD AND LLVIP

Dataset	Methods	Input size	P(%)	R(%)	mAP@0.5(%)	mAP@0.5:0.95(%)
AI-TOD	YOLOv5	640×640	57.0	38.4	39.4	17.0
	YOLOv6	640×640	**71.7**	31.1	33.7	14.9
	YOLOv8	640×640	52.1	41.9	41.0	18.0
	YOLOv8-G	640×640	48.2	38.5	37.4	16.2
	YOLOv9	640×640	51.6	41.2	40.7	17.9
	YOLOv10	640×640	45.4	41.6	39.5	17.8
	YOLOv11	640×640	52.5	42.1	41.4	17.9
	WL-RANet	640×640	54.4	**45.0**	**43.1**	**19.1**
LLVIP	YOLOv5	640×640	91.4	82.8	89.7	**52.3**
	YOLOv6	640×640	91.5	82.3	89.2	52.0
	YOLOv8	640×640	90.3	80.2	89.3	51.8
	YOLOv8-G	640×640	89.9	79.1	86.8	49.4
	YOLOv9	640×640	90.9	83.0	89.4	51.8
	YOLOv10	640×640	90.9	82.8	89.3	52.2
	YOLOv11	640×640	90.8	82.6	89.0	50.9
	WL-RANet	640×640	**91.8**	**84.2**	**89.8**	51.4

and advancement of our proposed model is further verified.

Finally, to validate the real-time performance of the proposed model, we conduct performance tests on the VisDrone 2019 dataset. The experimental results show that the WL-RANet model achieves a real-time processing frame rate of approximately 140 frames/s (FPS). Specifically, in the model's single-frame processing workflow, the preprocessing takes 0.1 ms, the inference process takes 3.4 ms, and the postprocessing stage takes 3.6 ms. The time distribution of each stage is reasonable, and the total processing time is controlled within 7.1 ms.

In the dynamic video scenario, we test using a drone video sequence containing 9910 frames. By performing frame-by-frame inference on the complete video sequence, the model achieves an average frame rate of 103.29 FPS in dynamic scenarios, further confirming its real-time performance. These experimental results fully demonstrate the real-time advantages of the proposed method in drone-based object detection tasks.

D. Ablation Experiment

We experimentally compare the proposed approach with other YOLO series models and verify the effectiveness of the algorithm through ablation experiments. YOLOv11 is chosen as the benchmark, considering the actual usage scenario of the model and the SOTA model in object detection. In addition, we choose the VisDrone 2019 dataset as a validation of model validity for the ablation experiment part. The experiments primarily include validating the effectiveness of the multichannel recursive residual fusion mechanism in the feature extraction network, evaluating the performance of the dynamic multichannel weight-learnable recursive aggregation network in the feature fusion process, testing the overall effectiveness of the WL-RANet network, and conducting comparative experiments with benchmark models and mainstream algorithms.

As shown in Table VII, the data highlighted in red indicate the optimal values, while those marked in blue correspond to the second-best values. The experimental results demonstrate that our proposed model performs well at each stage of small-object detection in complex scenarios for AAV applications.

In the ablation experiments, the YOLOv5–YOLOv11 serve as comparison models, without the incorporation of any additional mechanisms or network structures. As evident from the data in Table. VII, the performance metrics of these models (e.g., mAP@0.5 and mAP@0.5:0.95) are relatively lower compared with the proposed model, particularly with mAP@0.5:0.95 values ranging from 16.5% to 23.4%. Among these, the YOLOv9 model achieves 38.9% in mAP@0.5 and notably 23.4% in mAP@0.5:0.95, demonstrating the best performance among the comparison models.

First, we conduct a comparative experiment with the McWL network. The experimental results show that, compared to the YOLOv11 model, the McWL network improves mAP@0.5 by 1.0%, increases recall by 0.2%, and enhances precision by 2.9%. When compared with other models, the mAP@0.5 value increases from 38.9% (YOLOv9) to 39.4%, and the mAP@0.5:0.95 metric improves from 23.4% to 23.7%. These results indicate that the McWL network effectively addresses issues related to scale variation and multidimensional information dependency during the feature fusion process, thereby enhancing model performance.

Second, we evaluate the role of the McRF mechanism in the feature extraction network. Experimental data show that, compared with the YOLOv11 model, the introduction of the McRF mechanism improves the model's mAP@0.5 by 0.7%, increases recall by 0.6% and enhances precision by 2.5%. When compared with other models, mAP@0.5 increases from 38.9% (YOLOv9) to 39.1%, and mAP@0.5:0.95 improves from 23.4% to 23.5%. These results validate the effectiveness of the McRF mechanism in the feature extraction network, demonstrating its ability to help the network capture remote key information of small targets, although the improvement is slightly less significant than that of the McWL network.

Furthermore, we further validate the performance of the WL-RANet model by simultaneously incorporating both the McWL network and the McRF mechanism. Experimental results demonstrate that, compared to using the McWL network or McRF mechanism individually, WL-RANet achieves further improvements of 40.5% in mAP@0.5 and 24.3% in

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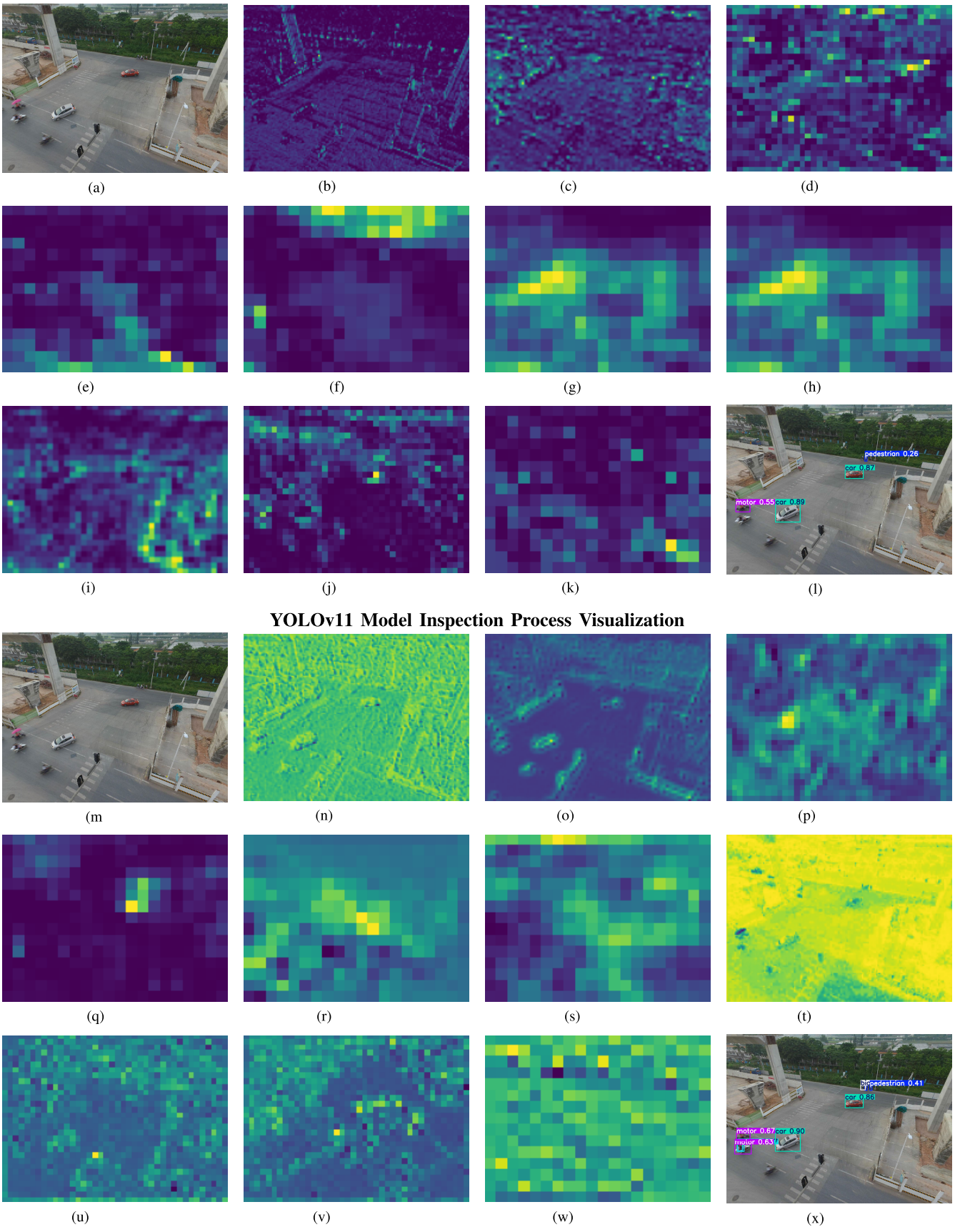


Fig. 10. Comparison of YOLOv11 and WL-RANet detection model detection process visualization results. (a)–(l) Results of feature map visualization for each stage of the baseline model. (m)–(x) Results of feature map visualization for each stage of the proposed model. (a) Input. (b) C3k2 d1. (c) C3k2 d2. (d) C3k2 d3. (e) C3k2 d4. (f) SPPF. (g) C2PSA. (h) FPN d1. (i) FPN d2. (j) PAN d1. (k) PAN d2. (l) Output. (m) Iuput. (n) McRF d1. (o) McRF d2. (p) McRF d3. (q) SPPF. (r) McRF d4. (s) C2PSA. (t) McWL d1. (u) McWL d2. (v) McWL d3. (w) McWL d3. (x) Output.

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TABLE VII
COMPARATIVE STUDY OF ABLATION EXPERIMENTS ON VISDRONE 2019

Model	McRF mechanism	McWL network	P	R	mAP@0.5(%)	mAP@0.5:0.95(%)
YOLOv5	-	-	49.5	36.3	37.1	22.1
YOLOv6	-	-	47.4	35.2	35.9	16.5
YOLOv8	-	-	48.3	37.8	38.0	22.9
YOLOv8-G	-	-	45.4	34.0	34.2	19.9
YOLOv9	-	-	50.0	38.3	38.9	23.4
YOLOv10	-	-	48.9	37.9	38.3	22.9
YOLOv11	-	-	48.5	37.8	38.4	22.9
McWL network	-	✓	51.4	38.0	39.4	23.7
McRF mechanism	✓	-	51.0	38.4	39.1	23.5
WL-RANet	✓	✓	51.1	40.0	40.5	24.3

TABLE VIII
MCWL IN EACH SOTA MODEL ON VISDRONE 2019

Model	McWL	P	R	mAP@0.5(%)	mAP@0.5:0.95(%)
YOLOv5	-	49.5	36.3	37.1	22.1
YOLOv6[arXiv2022]	-	47.4	35.2	35.9	16.5
YOLOv8[arXiv2024]	-	48.3	37.8	38.0	22.9
YOLOv8-G	-	45.4	34.0	34.2	19.9
YOLOv9[arXiv2024]	-	50.0	38.3	38.9	23.4
YOLOv10[arXiv2024]	-	48.9	37.9	38.3	22.9
YOLOv11[arXiv2024]	-	48.5	37.8	38.4	22.9
v5-McWL	✓	49.3 ↓	38.3 ↑	39.2 ↑	23.4 ↑
v6-McWL	✓	49.4 ↑	36.6 ↑	37.5 ↑	22.3 ↑
v8-McWL	✓	49.8 ↑	38.6 ↑	39.2 ↑	23.6 ↑
v8G-McWL	✓	47.1 ↑	37.8 ↑	36.6 ↑	21.4 ↑
v9-McWL	✓	52.0 ↑	39.4 ↑	40.4 ↑	24.2 ↑
v10-McWL	✓	49.3 ↑	37.8 ↓	39.0 ↑	23.5 ↑
v11-McWL	✓	51.4 ↑	38.0 ↑	39.4 ↑	23.7 ↑

mAP@0.5:0.95. When compared with the YOLOv11 model, WL-RANet exhibits a 2.1% increase in mAP@0.5, a 2.2% enhancement in recall, and a 2.6% improvement in precision. These results fully demonstrate the synergistic effects of combining the McWL network and McRF mechanism, which significantly elevate overall model performance. In addition, this validates that our proposed model effectively addresses the challenge of reduced detection accuracy for small objects in complex remote sensing imagery.

Through comparative analysis of ablation experiments, we have demonstrated that both the McWL network and the McRF mechanism positively impact model performance. The outstanding performance of WL-RANet further indicates that the combination of these two methods effectively enhances the model's detection accuracy, particularly in small-object detection tasks within AAV scenarios.

E. McWL Network Boost at SOTA Methods

In addition, we also validate the generalization for the McWL network, and the experimental results are shown in Table VIII. First, we choose as benchmarks the mainstream algorithms and SOTA models in recent years, which all have feature fusion networks combining FPN and PAN. We replace the PAN network with the feature fusion approach of McWL. With Table VIII, it is easy to see that all of our proposal McWL network plays a role in the benchmark network.

Compared with YOLOv5, v5-McWL has a 2.1% improvement in mAP@0.5 and a 2.0% increase in recall, but a slight 0.2% decrease in precision. In contrast to YOLOv6, v6-McWL has a 1.6% improvement in mAP@0.5 and a 1.4% increase in recall, along with a 2.0% improvement in precision. The v8-McWL has a 1.2% increase in mAP@0.5, a 0.8% increase in recall, and a 1.5% increase in precision compared with the YOLOv8. The v8g-McWL shows the most significant improvement over the YOLOv8-Ghost, with a 2.4% increase in mAP@0.5, a 3.8% increase in recall, and a 1.7% increase in precision. Compared to YOLOv9, v9-McWL offers a 1.5% improvement in mAP@0.5, a 1.1% increase in recall, and a 2.0% increase in precision. Upon comparison with YOLOv10, v10-McWL has a 0.7% improvement in mAP@0.5, a slight

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0.1% decrease in the recall, but a 0.4% improvement in precision. Finally, versus YOLOv11, v11-McWL shows a 1.0% improvement in mAP@0.5, a 0.2% increase in recall, and a significant 2.9% improvement in precision. The data analysis demonstrates that our proposed multichannel weights learnable recursive aggregation network is general and can be well migrated to other feature fusion network architectures.

V. Conclusion

In this article, we propose the weights learnable recursive aggregation detection framework for AAV remote sensing image scenarios. Specifically, we propose a generalized feature fusion network (McWL) and a plug-and-play representation extraction mechanism (McRF). The McWL network adopts the recursive idea of dynamically weighting different layers of scale features, and at the same time, recursively aggregates the features of different layers. In addition, we verify the generalization of the McWL network by transferring its idea to other object detection network models. The McRF mechanism is designed to extract key representations from sparse features. It features a multibranch structure, with its core comprising a recursive fusion bottleneck and an efficient multibranch feature extraction module. These components enable the extraction of feature representations at different depths, enrich object information, and thereby enhance the network's ability to detect objects in AAV images.

We experimentally validate our methods, which show better performance on the VisDrone 2019, AI-TOD, and LLVIP datasets. The overall performance of the network is first validated and compared between the SOTA and mainstream algorithms, and the WL-RANet framework achieves accuracies of 40.5%, 43.1%, and 89.8% (mAP@0.5) on the VisDrone 2019, AI-TOD, and LLVIP datasets, exceeding the given SOTA method. Subsequently, we migrate the McWL network to various mainstream methods to evaluate its generalization capability. The experimental results show that McWL contributes to the feature fusion capability of different networks, and the effect is especially more significant in the YOLOv5 and YOLOv8-Ghost models.

Although the WL-RANet network has achieved good results in AAV image object detection tasks and may be applied to real-time urban UAV detection processing in the future, it still has some limitations.

1) The AAV obtains the object image from a high-altitude overhead shot and therefore only obtains information related to the top view of the object, which causes the network to misdetect phenomena such as the top view of cars and vans.
2) Our method only performs object detection in AAV optical images, which is highly susceptible to degradation of detection accuracy under conditions of contaminated object appearance (light and raindrops).

Therefore our method still needs further research and optimization. In the future, we may enhance the AAV image object detection performance by multisource image synergy.

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